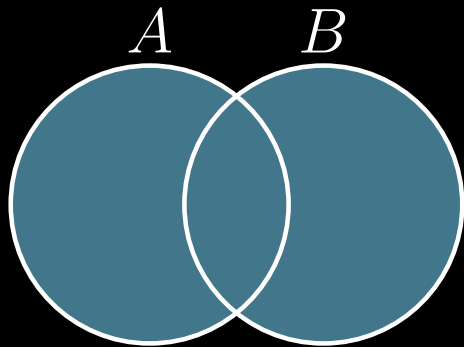


Union and Intersection of Sets

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The **union of two sets**, written as $A \cup B$, is the set containing all elements that belong to A , B , or both.

Union Diagram



$$A \cup B$$

Union Example

$$A = \{1, 2, 3\}$$

$$B = \{3, 4, 5\}$$

$$A \cup B = \{1, 2, 3, 4, 5\}$$

Write Common Elements Once

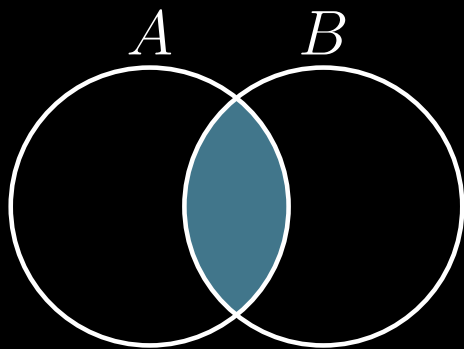
Elements that appear in both sets are written only once in the union.

$$A \cup B = \{1, 2, 3, 4, 5\}$$

The element 3 is in both sets, but it is written only **once**.

The **intersection of two sets**, written as $A \cap B$, is the set containing only the elements common to both sets.

Intersection Diagram



$$A \cap B$$

Intersection Example

$$A = \{1, 2, 3\} \quad B = \{3, 4, 5\}$$

$$A \cap B = \{3\}$$

Only 3 is in both sets.

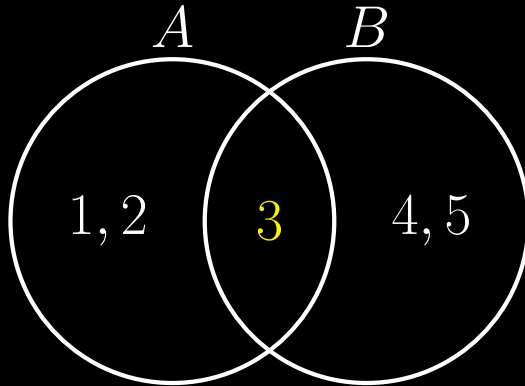
No Common Elements

If two sets have no common elements, then their intersection is the **empty set**, written as $\emptyset$.

$$\{1, 2\} \cap \{5, 6\} = \emptyset$$

Venn Diagrams

Venn diagrams help visualize the union and intersection of sets.



Reading the Venn Diagram

Everything inside **either** circle is the union.

$$A \cup B = \{1, 2, 3, 4, 5\}$$

The **overlap** is the intersection.

$$A \cap B = \{3\}$$

Union vs. Intersection

Symbol	Name	Meaning
$A \cup B$	Union	in A or B
$A \cap B$	Intersection	in A and B

Memory Tip

$\cup$ looks like the letter **U**, for **U**nion.

Union = join everything (**or**).

Intersection = keep only what is shared (**and**).

Steps

Union

1. List every element of A and B .
2. Write repeated elements once.

Intersection

1. Find the elements found in both sets.

Examples

Example 1

Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{4, 5, 6, 7\}$.
Find $A \cup B$ and $A \cap B$.

$$A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$$

$$A \cap B = \{4, 5\}$$

Example 2

Let $M = \{b, c, d\}$ and $N = \{d, e\}$. Find $M \cup N$ and $M \cap N$.

$$M \cup N = \{b, c, d, e\}$$

$$M \cap N = \{d\}$$

Example 3

Let $P = \{3, 6, 9\}$ and $Q = \{2, 4, 8\}$. Find $P \cup Q$ and $P \cap Q$.

$$P \cup Q = \{2, 3, 4, 6, 8, 9\}$$

$$P \cap Q = \emptyset$$

The sets have no common elements.

Example 4

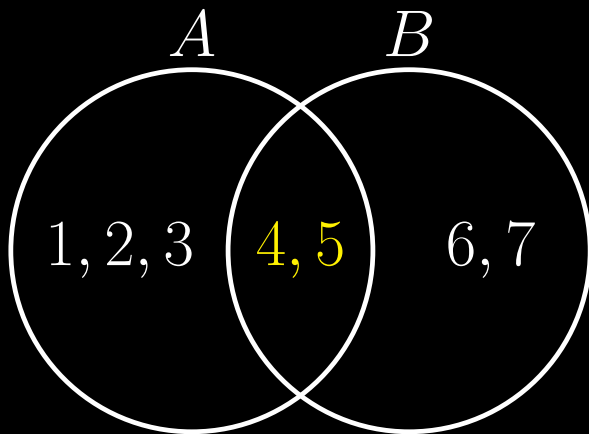
In a class survey, $A = \{\text{Mia, Noel, Olga}\}$ like soccer and $B = \{\text{Olga, Paul, Quinn}\}$ like chess. Find $A \cup B$ and $A \cap B$.

$$A \cup B = \{\text{Mia, Noel, Olga, Paul, Quinn}\}$$

$$A \cap B = \{\text{Olga}\}$$

Example 5

Venn diagram for Example 1:



Practice Exercises

Practice 1

Let $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$. Find $A \cup B$ and $A \cap B$.

Practice 2

Let $M = \{a, e, i, o\}$ and $N = \{i, o, u\}$. Find $M \cup N$ and $M \cap N$.

Practice 3

Let $P = \{2, 4, 6, 8\}$ and $Q = \{1, 3, 5, 7\}$. Find $P \cup Q$ and $P \cap Q$.

Practice 4

Let $X = \{5, 10, 15, 20\}$ and $Y = \{10, 20, 30\}$.

Find $X \cup Y$ and $X \cap Y$.

Practice 5

Let $A = \{m, n, p, q\}$ and $B = \{p, q, r, s\}$.

Find $A \cup B$ and $A \cap B$.

Practice 6

Let $C = \{7, 8, 9, 10\}$ and $D = \{8, 10, 12\}$.

Find $C \cup D$ and $C \cap D$.

Practice 7

Let $R = \{2, 3, 5, 7, 11\}$ and $S = \{5, 7, 13\}$.

Find $R \cup S$ and $R \cap S$.

Practice 8

Let $E = \{x, y, z\}$ and $F = \{y, z, w\}$. Find $E \cup F$ and $E \cap F$.

Practice 9

Let $G = \{12, 14, 16\}$ and $H = \{14, 18, 20\}$.

Find $G \cup H$ and $G \cap H$.

Practice 10

In a class survey, $A = \{\text{Ana, Ben, Carl, Dana}\}$
like basketball and $B =$
 $\{\text{Carl, Dana, Ella, Fred}\}$ like volleyball.
Find $A \cup B$ and $A \cap B$.

Activities

Activity 1

Let $A = \{2, 4, 6, 8\}$ and $B = \{6, 8, 10, 12\}$.

Find $A \cup B$ and $A \cap B$.

Activity 2

Let $M = \{a, b, c, d\}$ and $N = \{c, d, e, f\}$.

Find $M \cup N$ and $M \cap N$.

Activity 3

Let $P = \{9, 11, 13\}$ and $Q = \{2, 4, 6, 8\}$. Find $P \cup Q$ and $P \cap Q$.

Activity 4

Let $X = \{15, 20, 25\}$ and $Y = \{20, 30, 35\}$.
Find $X \cup Y$ and $X \cap Y$.

Activity 5

Let $A = \{h, i, j, k\}$ and $B = \{j, k, l, m\}$.

Find $A \cup B$ and $A \cap B$.

Activity 6

Let $C = \{14, 16, 18, 20\}$ and $D = \{16, 18, 22\}$. Find $C \cup D$ and $C \cap D$.

Activity 7

Let $R = \{1, 2, 3, 5, 8\}$ and $S = \{3, 5, 13\}$.

Find $R \cup S$ and $R \cap S$.

Activity 8

Let $E = \{r, s, t\}$ and $F = \{t, u, v\}$. Find $E \cup F$ and $E \cap F$.

Activity 9

Let $G = \{21, 24, 27\}$ and $H = \{24, 27, 30\}$.
Find $G \cup H$ and $G \cap H$.

Activity 10

In a class survey, $A = \{\text{Grace, Henry, Ivan, Joy}\}$ like reading and $B = \{\text{Ivan, Joy, Kate, Liam}\}$ like drawing. Find $A \cup B$ and $A \cap B$.

Thank you!